

AI Powered TDS Deduction Issue Detection System

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Approvals

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1. Introduction

1.1 Project Summary

1.1.1 Objective

The objective of this project is to design and develop an AI-powered TDS (Tax Deducted at Source) Deduction Issue Detection System that automatically identifies, flags, and reports common errors and non-compliances occurring during TDS deduction processes across applicable sections of the Income Tax Act, 1961.

The system aims to reduce manual effort, eliminate human error, ensure regulatory compliance, and provide real-time alerts to finance and tax teams when TDS deductions are incorrectly applied, missed, or miscalculated.

1.1.2 Background

TDS (Tax Deducted at Source) is a mechanism under the Indian Income Tax Act, 1961, by which tax is collected at the source of income. Organizations acting as deductors are required to deduct TDS while making specified payments such as salary, interest, rent, professional fees, contract payments, and more.

Despite being a well-established compliance process, TDS deduction in practice is prone to numerous errors — from selecting the wrong section, applying incorrect rates, missing threshold tracking, incorrect handling of GST components, to misclassifying vendor residency status. These errors result in interest liabilities under Section 201(1A), penalties, and scrutiny from tax authorities.

Currently, most organizations handle TDS deduction through manual processes or ERPs that are not adequately configured, leading to persistent compliance gaps. There is a clear need for an intelligent system that can proactively detect these issues before they result in **regulatory** consequences.

Regulatory Update — Effective 1st April 2026: Section 192 (Salary) is renamed to Section 392, and Section 192A (Other than Salary) is renamed to Section 393. The system must reflect these updated section numbers in all validations, reports, dashboards, and alerts from 1st April 2026 onwards.

1.2 Project Scope

1.2.1 In-Scope Functionalities

The system shall cover the detection of TDS-related issues across the following areas:

A. Section Applicability & Selection — The system shall validate whether the correct TDS section has been applied for a given payment type. This includes detection of misclassification between commonly confused sections such as 194C vs. 194J (contract vs. professional fees) and 194H vs. 194C (commission vs. contract payments). The system will cover the following TDS sections:

- Section 193 — Interest on Securities
- Section 194 — Dividend
- Section 194A — Interest (Other than Securities)
- Section 194C — Contract Payments
- Section 194H — Commission / Brokerage
- Section 194I — Rent

- Section 194J — Professional / Technical Fees
- Section 194K — Income from Mutual Funds
- Section 194IA — Purchase of Immovable Property
- Section 194LA — Land Acquisition Compensation
- Section 194N — Cash Withdrawal
- Section 194O — E-commerce Transactions
- Section 194Q — Purchase of Goods
- Section 195 — Non-Resident Payments
- Sections 196A / 196B / 196C / 196D — Non-Resident Specific Incomes
- Section 392 — Salary (renamed from Section 192, effective 1st April 2026)
- Section 393 — Other than Salary (renamed from Section 192A, effective 1st April 2026)

B. TDS Rate Validation — The system shall validate the rate of TDS applied for each transaction, including detection of: incorrect rates under a correctly identified section, failure to apply the higher 20% rate under Section 206AA when PAN is unavailable, failure to apply the higher rate under Section 206AB for non-filers, non-application of lower or nil deduction certificates (Form 197), and failure to apply applicable DTAA rates for non-resident payments.

C. Threshold & Limit Tracking — The system shall track payment thresholds on a vendor-wise and section-wise basis to detect: TDS being deducted when the threshold has not been crossed, TDS not being deducted after the threshold has been crossed, incorrect aggregation of payments across branches for the same vendor, and incorrect aggregation per invoice instead of per PAN.

D. Timing of Deduction — The system shall validate that TDS is deducted at the correct trigger point — i.e., at the time of credit or payment, whichever is earlier — to detect late deductions that attract interest under Section 201(1A), TDS missed on advance payments, and TDS not deducted on year-end provision entries.

E. Nature of Payment Classification — The system shall analyze payment descriptions and contract types to detect misclassification errors including: incorrect splitting of composite contracts (material + service), wrongful exclusion of reimbursements from TDS base, confusion between discounts and commission, and misclassification of salary vs. professional fee payments.

F. Vendor Master Data Validation — The system shall validate vendor master data to detect: missing or incorrect PAN, duplicate vendor entries leading to incorrect threshold calculation, undefined vendor category (individual / company / firm / HUF), incorrect residential status, and absence of transporter tagging for 194C exemption.

G. TDS Calculation Accuracy — The system shall verify TDS calculation logic to detect: gross vs. net base confusion, incorrect deduction on GST components, rounding errors, and incorrect proportional deduction on partial payments.

H. Section-Wise Specific Issue Detection — The system shall perform granular, section-specific validations for high-risk sections including but not limited to 194A, 194C, 194H, 194I, 194J, 194IA, 194Q, 194O, 195, 194N, 194LA, 392, and 393, each covering the specific mistakes documented in the source requirements.

I. Cross-Section Common Issue Detection — The system shall detect cross-cutting issues applicable across all sections, including: PAN missing or invalid leading to non-application of 20% rate, 206AB non-filer higher rate not applied, timing errors (wrong deduction point, missed provisions), GST component incorrectly included in TDS base, and threshold not aggregated vendor-wise.

J. AI-Driven Compliance Intelligence & Risk Analysis - The system will leverage historical transaction data to build vendor behavior profiles, detect anomalies, and identify recurring compliance patterns and risks. It will also use intelligent inference to recover missing critical data (e.g., PAN) and generate risk scores to highlight high-risk transactions, vendors, and organizational areas for proactive compliance management.

2. System Perspective

2.1 Assumptions

The following assumptions have been made in defining the requirements of this system:

- The system will integrate with the organization's existing ERP or accounting software to receive transactional payment data in a defined format.
- PAN data for all vendors will be available in the vendor master and will be accessible to the system for validation.
- The system will have access to the current Income Tax Act provisions, applicable TDS rates, and section-wise threshold limits, which will be maintained and updated by the system administrator.
- DTAA rate tables for applicable countries will be maintained as reference data within the system.
- Form 15G/15H submission records and lower deduction certificates (Form 197) will be made available to the system to suppress TDS flags where exemptions are legitimately claimed.
- Vendor residential status (resident vs. non-resident) will be maintained in the vendor master and will be validated by the system.
- Year-end provision data will be available as input to the system for TDS deduction validation.
- The system will be used by finance, accounts payable, and tax compliance teams within the organization.
- The system will not directly process tax payments or file returns but will flag issues for human review and action.

- Branch-wise payment data will be consolidated at the organization level to support vendor-wise threshold tracking.
- The system will recognise Section 392 (Salary) and Section 393 (Other than Salary) as the updated section numbers effective 1st April 2026, replacing the former Section 192 and Section 192A respectively.
- The system will maintain historical transaction data required for building vendor behavior profiles and pattern analysis.
- Sufficient historical data will be available to support anomaly detection and risk scoring models.

2.2 Constraints

- The system scope is limited to TDS sections listed under Section 1.2.1. Sections explicitly excluded — including 194B (Lottery), 194BA (Online Gaming), 194BB (Horse Race), 194D (Insurance Commission), 194R (Benefit/Perquisite), 194S (VDA/Crypto), 194LBA/LBB/LBC, 194G, 194DA, 194EE, 194F, 194IB, 194IC, 194M, and 194P — are out of scope for this release.
- The system must comply with the provisions of the Income Tax Act, 1961 as amended and in force at the time of implementation; any legal changes post-go-live will require a configuration update.
- Data accuracy of detection is contingent on the quality and completeness of vendor master data, PAN records, and transaction data provided as input.
- The system will not override or auto-correct transactions; it will only flag and report detected issues for human review.
- The system must be capable of handling multi-branch, multi-entity data aggregation for threshold tracking.
- Performance of the system must not adversely impact the speed of the underlying ERP or payment processing workflows.
- All data processed by the system must comply with applicable data privacy and information security policies of the organization.
- Accuracy of AI-based insights is dependent on the availability and quality of historical transaction data and user feedback.
- Initial system performance may rely more on rule-based detection until sufficient historical data is accumulated for AI models.

3. Business Requirements

3.1 Process Flow

The TDS Issue Detection System shall operate through the following high-level process flow:

- **Step 1 — Transaction Intake:** Payment transactions are received from the ERP / accounting system either in real-time (on-transaction) or in batch mode (periodic file upload). Each transaction record shall include: payment amount, payment date, vendor/payee details, PAN, vendor category, residential status, nature of payment, applicable TDS section (as tagged in ERP), TDS amount deducted, and GST components.
- **Step 2 — Vendor Master Validation:** The system validates vendor master records against each transaction to detect: missing or invalid PAN, incorrect vendor category, wrong residential status, missing transporter flag, and duplicate vendor entries.
- **Step 3 — Section Applicability Check:** The system validates whether the TDS section applied to the transaction matches the nature of payment. Misclassifications are flagged with a recommended correct section and applicable rate. From 1st April 2026, the system validates the correct use of Section 392 (Salary) and Section 393 (Other than Salary) in place of former Sections 192 and 192A.
- **Step 4 — Rate Validation:** The system checks the TDS rate applied against the correct rate for the identified section, vendor category, PAN availability status, and non-filer status (206AB). DTAA rates are validated for non-resident payments under Section 195. Applicable lower deduction certificates are checked before flagging rate errors.
- **Step 5 — Threshold Check:** The system aggregates all payments to the same vendor (PAN-wise) within the financial year to determine if the applicable section threshold has been crossed. Transactions below threshold but with TDS deducted, and transactions above threshold without TDS deducted, are flagged accordingly.
- **Step 6 — Timing Validation:** The system validates whether TDS was deducted at the correct trigger event (credit or payment, whichever is earlier). Advance payments and year-end provisions are specifically checked for missed TDS deductions.
- **Step 7 — Calculation Verification:** The system recalculates the correct TDS amount based on the validated section, rate, and taxable base (excluding GST where applicable) and compares it against the actual TDS deducted. Discrepancies beyond defined tolerance are flagged.

- **Step 8 — Issue Reporting:** All flagged issues are compiled into a structured issue report categorised by section, issue type, severity (high / medium / low), and transaction reference. The report is made available to the finance/tax team for review and corrective action.
- **Step 9 – AI-Driven Insights:** The system will analyze transactions against historical vendor behavior and patterns to detect anomalies, misclassifications, and recurring compliance issues. It will assign risk scores to prioritize high-risk transactions and include AI-driven insights such as anomaly indicators and behavioral deviations in the final issue reports.

- **Step 10 - Correction / Reversal Transaction Handling (New Step in Process Flow)**

The system shall detect transactions where a TDS error was already corrected via a **reversal entry** in the ERP. For such corrected transactions, the system shall report: original document, reversal document, correction document, wrong TDS amount, correct TDS amount, and shortfall — rather than flagging them as open issues.

3.2 User Interface Requirements

- **Dashboard:** The system shall provide a summary dashboard displaying: total transactions processed, count and category-wise breakdown of issues flagged, section-wise compliance health indicators, pending issues requiring action, and trend analysis over selected time periods. The dashboard shall also display AI-driven insights such as vendor risk scores, anomaly trends.
- **Issue Review Screen:** A transaction-level issue review screen shall display each flagged transaction with: transaction details (vendor, amount, date, section applied, TDS deducted), the nature of the detected issue, recommended corrective action, and status (open / acknowledged / resolved). Each issue shall include AI-generated insights such as anomaly flags, and suggested corrections based on historical patterns.
- **Vendor Master View:** A vendor master view shall allow authorized users to review and update vendor data including PAN, vendor category, residential status, transporter flag, and lower deduction certificate

details. The system shall display inferred or suggested data (e.g., PAN) along with confidence scores where applicable.

- **Threshold Tracker:** A threshold tracking module shall display real-time or near-real-time cumulative payment totals per vendor per section for the current financial year, with visual indicators as payments approach or cross applicable thresholds.
- **Reports & Exports:** The system shall support export of issue reports and threshold summaries in standard formats (Excel / PDF / CSV) for use in compliance reviews, audits, and filing support.
- **Risk & Analytics View:** The system shall provide a dedicated view for analyzing compliance trends, vendor risk scores, and recurring issue patterns across time periods.
- **Notifications & Alerts:** The system shall support configurable email or in-application alerts triggered on: high-severity TDS issues detected, threshold limits approaching or breached, and transactions flagged for non-deduction.

4. Appendices

4.1 List of Acronyms

Acronym	Full Form
TDS	Tax Deducted at Source
BRD	Business Requirements Document
PAN	Permanent Account Number
DTAA	Double Tax Avoidance Agreement
ERP	Enterprise Resource Planning
HUF	Hindu Undivided Family
NBFC	Non-Banking Financial Company
VDA	Virtual Digital Asset
AIF	Alternative Investment Fund
NSS	National Savings Scheme

UTI	Unit Trust of India
GST	Goods and Services Tax
UI	User Interface

4.2 Glossary

- **Deductor** — The person or entity responsible for deducting TDS at the time of making a specified payment.
- **Deductee** — The person or entity from whose payment TDS is deducted.
- **TDS Certificate** — A certificate issued by the deductor to the deductee acknowledging the TDS deducted and deposited (Form 16 / Form 16A).
- **Section 206AA** — Provision requiring deduction of TDS at 20% if the deductee fails to furnish their PAN.
- **Section 206AB** — Provision requiring deduction of TDS at a higher rate for specified persons who have not filed income tax returns for the preceding two years.
- **Section 392 (formerly Section 192)** — TDS on Salary payments, renamed effective 1st April 2026 as per the latest amendment to the Income Tax Act, 1961.
- **Section 393 (formerly Section 192A)** — TDS on Other than Salary payments, renamed effective 1st April 2026 as per the latest amendment to the Income Tax Act, 1961.
- **Form 15G / 15H** — Self-declaration forms submitted by a deductee to claim exemption from TDS deduction (15G for non-senior citizens, 15H for senior citizens), subject to eligibility conditions.
- **Form 197** — Certificate issued by the Income Tax Department authorising lower or nil TDS deduction for a deductee.
- **Form 26QB** — TDS return-cum-challan form for reporting TDS deducted on purchase of immovable property under Section 194IA.
- **Form 15CA / 15CB** — Documents required for foreign remittances; 15CA is filed by the remitter and 15CB is a chartered accountant certificate, applicable for payments under Section 195.
- **Threshold Limit** — The minimum payment amount below which TDS deduction is not applicable for a given section in a financial year.

- **Composite Contract** — A contract that includes both supply of goods/materials and provision of services, requiring careful classification for TDS purposes.
- **Provision Entry** — An accounting entry made at year-end to record an expense that has been incurred but not yet paid, which may trigger TDS deduction obligations.
- **Residential Status** — The classification of a taxpayer as a resident or non-resident under the Income Tax Act, which determines the applicable TDS provisions and rates.

4.3 Related Documents

Document	Description
Income Tax Act, 1961	Primary legislation governing TDS provisions in India
CBDT Circulars & Notifications	Clarifications and amendments issued by the Central Board of Direct Taxes
Form 26Q / 27Q / 26QB / 27EQ	Quarterly TDS return forms for various payment categories
Form 16A	TDS certificate for non-salary deductions
DTAA Treaties	Bilateral tax treaties between India and other countries applicable to Section 195
Vendor Onboarding Policy	Internal policy governing vendor PAN collection and category classification
ERP TDS Configuration Guide	Technical guide for TDS section and rate configuration in the ERP system
Raw Requirements Document	Source document: Common Issues in TDS Deduction (India) — Full Spectrum